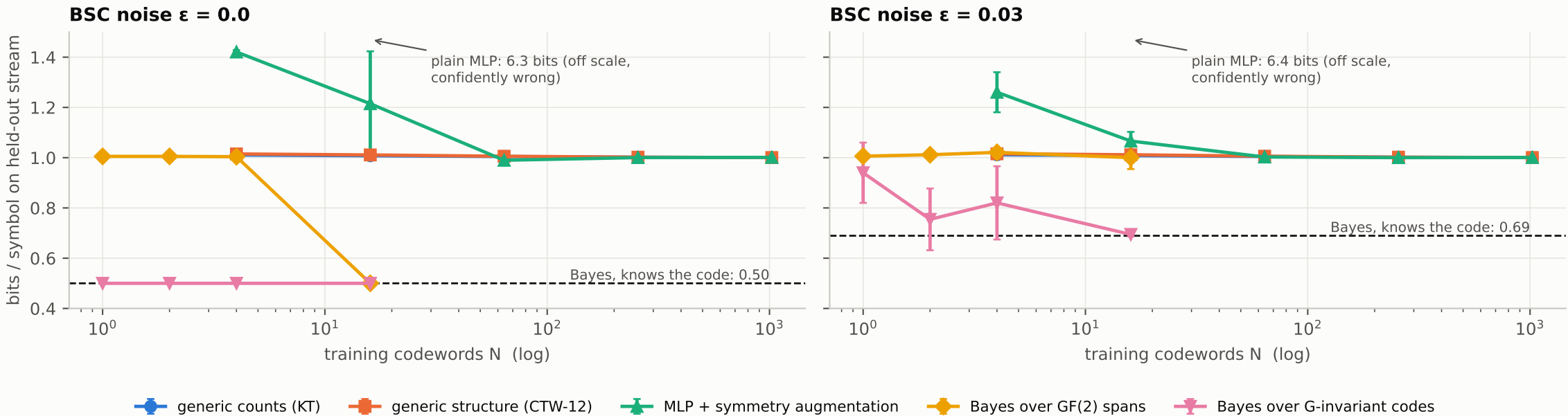


Fig. 4 · A sporadic symmetry earns bits — but only through a learner that can represent it



E002 + E003. Extended Golay code $[24,12,8]$, $G = \text{PSL}(2,23) \subset M_{24}$ (order 6072, verified; transitive on the 759 octads). Bayes over G-invariant linear codes reaches the optimum 0.500 from ONE codeword; linearity alone needs 16; SGD (even with orbit augmentation) extracts ~nothing — the check bits are parities. Error bars: sd over seeds (2–5).